TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Estimation of Ferrous Ions by Standard Potassium Permanganate

The 1 : 5 permanganate–iron(II) titration — no indicator, a pale-green solution, a yellow product that tries to hide the end point, and the first blush of pink that finishes it.

CHE-1206 · CHEMISTRY-II SESSIONAL · EXPERIMENT 8 · LEVEL 1 TERM 2

1. Aim and theory

Aim: to determine the amount of iron(II) in a supplied solution, using the permanganate standardised in Experiment 7. Given a solution made by dissolving 2.50 g of an iron-bearing sample in dilute sulphuric acid (all the iron brought into the +2 state) and making up to 250 mL, the titration reports the iron concentration — and with it the iron content of the sample. In acid solution permanganate oxidises iron(II) to iron(III) rapidly and completely, in the cold:

$$MnO_4^- + 5\ Fe^{2+} + 8\ H^+ \rightarrow Mn^{2+} + 5\ Fe^{3+} + 4\ H_2O \quad (1\ \text{mol permanganate} : 5\ \text{mol } Fe^{2+})$$

Note the ratio: **1 : 5**, not the 2 : 5 of the oxalate standardisation — each Fe^{2+} gives up one electron, each oxalate two. **No indicator is used**: permanganate is again self-indicating, and the end point is the first permanent pale pink. Two practical wrinkles matter. First, the product Fe^{3+} is yellow, and its colour can mask a faint pink; a little phosphoric acid ties Fe^{3+} up as a colourless phosphate complex and sharpens the end point (the idea behind the Zimmermann–Reinhardt reagent). Second, Fe^{2+} is slowly oxidised by air, so the solution is kept acidified, titrated promptly, and not left standing between repeats.

2. Apparatus and solutions

- Standard $KMnO_4$ solution, 0.0200 M (standardised in Experiment 7), in the burette.
- The supplied iron(II) solution (2.50 g of sample made up to 250 mL); pipette 25 mL.
- Dilute sulphuric acid (about 2 M); a few drops of phosphoric acid (optional, to bleach the Fe^{3+} colour); conical flask, white tile.

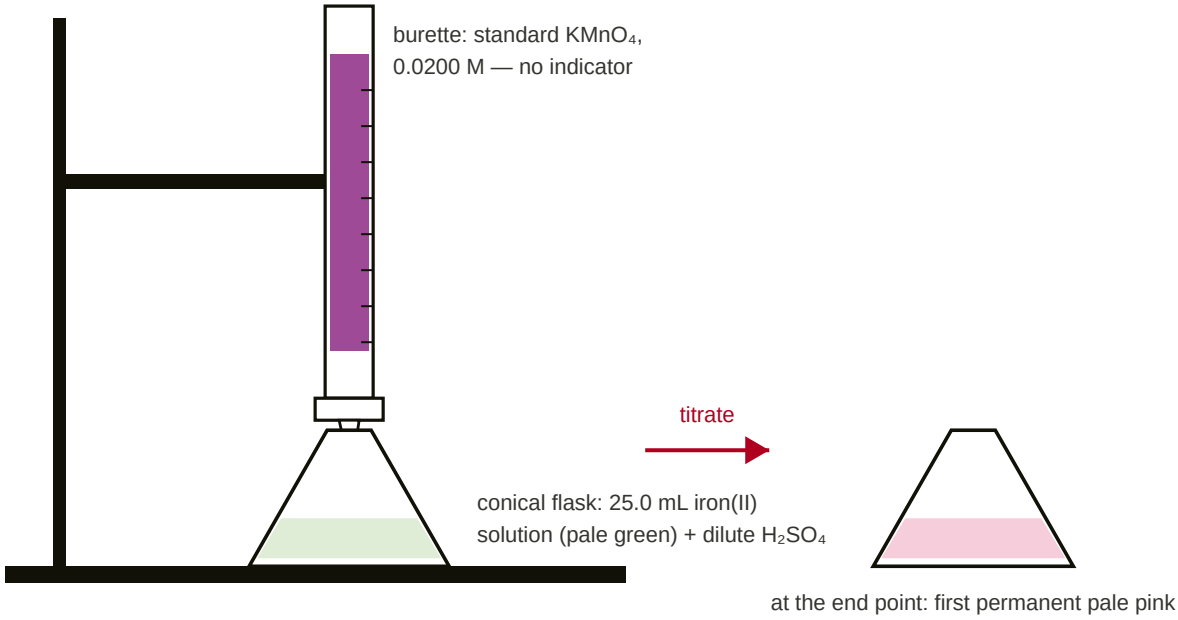


Figure 1 — Titrating iron(II) with standard permanganate. The pale-green Fe^{2+} solution takes the purple drops colourlessly until the iron is spent; the next drop stays pink.

3. Procedure

1. Rinse and fill the burette with the standard $KMnO_4$; read the upper meniscus.
2. Pipette 25.0 mL of the iron(II) solution into a conical flask; add about 20 mL of dilute sulphuric acid (and a few drops of phosphoric acid, if used, to keep the Fe^{3+} colour down).
3. Titrate in the cold with constant swirling — unlike the oxalate titration, no heating is needed; the reaction is fast at room temperature.
4. Slow to drops near the end; the end point is the first **pale pink that persists for 30 seconds**. (The pink fades slowly on standing as traces of reducing matter consume the excess drop — 30 seconds' persistence is the agreed criterion, not permanence forever.)
5. Repeat until two titres agree within 0.1 mL; average the concordant titres.

4. Worked calculation (model data)

ESTIMATION RESULT

Standard $KMnO_4$: 0.0200 M. Volume of iron solution pipetted: 25.0 mL. Concordant titre: 22.4 mL.

Moles of $KMnO_4 = 0.0200 \times 0.0224 = 4.48 \times 10^{-4}$ mol.

Moles of $Fe^{2+} = 5 \times 4.48 \times 10^{-4} = 2.24 \times 10^{-3}$ mol in the 25.0 mL pipette (the 1 : 5 ratio does all the work).

Concentration of $Fe^{2+} = 2.24 \times 10^{-3} / 0.0250 =$ **0.0896 M**.

Iron per litre = 0.0896 × 55.85 = **5.00 g L^{-1}**.
The 250 mL flask therefore holds 5.00 × 0.250 = 1.25 g of iron, from a 2.50 g sample: iron content = 1.25/2.50 × 100 = **50.0 %**.

5. Observation table (as recorded)

TRIAL	INITIAL READING (ML)	FINAL READING (ML)	TITRE (ML)
1 (rough)	0.0	22.9	22.9
2	0.0	22.4	22.4
3	0.0	22.4	22.4

Duplicates in volumetric work are not wasted effort: agreement between two independent pipette-and-titrate cycles is what turns a burette reading into a measurement.

6. Precautions

- Titrate promptly after acidifying: dissolved air oxidises Fe^{2+} on standing, and every ion lost to the air is an ion the burette never sees (result low).
- Keep the solution well acidified throughout — in weakly acid or neutral solution permanganate is reduced only to brown MnO_2 and the 1 : 5 stoichiometry collapses.
- Do not heat: the reaction is already fast in the cold, and hot acidified Fe^{2+} oxidises in air all the faster.
- Read the pink at 30 seconds' persistence; chasing a permanently unfading pink overshoots the end point, drop by drop.
- The pipette is rinsed with the iron solution, the burette with permanganate; the flask with distilled water only.

7. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why is no indicator used in this titration, and what exactly is the end point?
2. State the mole ratio of the reaction, and the wrong result a student gets by importing the 2 : 5 ratio from the oxalate experiment.
3. 25.0 mL of an iron(II) solution requires 18.6 mL of 0.0200 M $KMnO_4$. Find the iron concentration in g L^{-1}.

4. Why does the end-point pink fade if the flask is left for a few minutes, and why is that not a failed titration?
5. What does phosphoric acid do in this titration, and what problem does it solve?
6. Why must the iron solution be acidified and titrated without delay?

Answer key

1. Permanganate is self-indicating: its purple colour is discharged by Fe^{2+} as long as any remains, and the first excess drop — far too dilute to show in the burette's bulk — colours the whole flask pale pink. The end point is that first pink persisting 30 seconds.
2. 1 mol MnO_4^- : 5 mol Fe^{2+}, because $Fe^{2+} \rightarrow Fe^{3+}$ is a one-electron change. Using the oxalate ratio (2 : 5, i.e. 0.4 mol permanganate per mole of reductant instead of 0.2) doubles the permanganate credited per iron and halves the calculated iron — a factor-of-two error from a borrowed ratio.
3. Moles $KMnO_4 = 0.0200 \times 0.0186 = 3.72 \times 10^{-4}$; moles $Fe^{2+} = 5 \times 3.72 \times 10^{-4} = 1.86 \times 10^{-3}$ in 25.0 mL; mass $= 1.86 \times 10^{-3} \times 55.85 = 0.1039$ g per 25.0 mL $\rightarrow \times 40 = 4.16$ g L^{-1}.
4. The excess permanganate drop is slowly consumed by traces of reducing matter (dust, organics, the last Fe^{2+} diffusing from the flask wall), so its colour drains away. The titration was read at the agreed 30-second persistence, while the iron was exactly spent — slow fading afterwards changes nothing that was measured.
5. Phosphoric acid complexes Fe^{3+} into a colourless phosphate complex, removing the yellow colour that otherwise tints the solution and blurs the first faint pink of the end point.
6. Air oxidises Fe^{2+} to Fe^{3+}, faster in near-neutral solution; acid both slows that oxidation and guarantees the correct five-electron reduction of permanganate to Mn^{2+}. Delay between preparation and titration quietly converts analyte into product the burette cannot count.

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